GBU 2301: Business Statistics

Practice Questions: Chapters 1–4

Fall 2026 | Dr. Ömer ÇAYIRLI

Without solutions

Chapter 1: Data and Statistics

Questions 1–6.

Support each answer with the information in the case; definitions alone are not sufficient.

Question 1 (Easy)

Alpha Repairs records parts used and the elapsed completion time of each job. Each listed part was used once.

Job	Part used	Job completion time (minutes)
J01	Battery	30
J01	Cable	30
J01	Connector	30
J02	Screen	90

a. Construct a table suitable for reporting average completion time per job. State what one observation represents and explain the effect of averaging the original rows. No calculation is required.

b. Can the original register be used directly to count each type of part used? Explain.

Question 2 (Easy)

Beta Service Centres records these variables:

Variable	Example value	Meaning
centre_code	104	Identifier assigned to the centre
service_rating	4	Customer chooses one category: very poor = 1, poor = 2, fair = 3, good = 4, very good = 5
complaint_count	8	Number of complaints received that day
waiting_minutes	12.5	Measured customer waiting time
room_temperature_c	20	Room temperature in degrees Celsius

a. Classify each variable in a table: categorical or quantitative, discrete or continuous where applicable, and measurement scale.

b. Which statements below can be defended from the stated measurements? Explain each decision.

Statement	Comparison
A	A centre at 20°C is 5°C warmer than one at 15°C.
B	A rating of 4 indicates twice the service quality of a rating of 2.
C	A waiting time of 25 minutes is twice a waiting time of 12.5 minutes.
D	Centre 104 has higher service quality than Centre 102.

Question 3 (Average)

Gamma Retail records monthly sales at two kiosks:

Kiosk	Month	Sales (thousand MAD)
Alpha	January	40
Alpha	February	60
Alpha	March	80
Beta	January	80
Beta	February	60
Beta	March	40

Both kiosks have average monthly sales of 60 thousand MAD over these three months. An analyst proposes replacing the six records with those two averages for the following reports:

Report	Purpose
A	Compare total sales contributed by each kiosk during January–March
B	Compare the kiosks' sales in March
C	Examine each kiosk's sales pattern when deciding April stock allocations

a. For which reports would the two averages retain the information needed for the stated sales comparison? For the others, specify the records you would use and identify their data structure.

Question 3 (Average)

b. State what one observation represents in the original table and name the structure of that dataset.

c. Management proposes equal April stock allocations because the averages are equal. Give a recommendation supported by the available records.

Question 4 (Average)

Delta Transport wants to understand satisfaction among all individual passengers who used its intercity bus services during September.

Survey information	Detail
Invitation list	6,000 September passengers who booked through the mobile app
Invitation procedure	Every passenger on the list was invited once
Responses	1,200 passengers, with one response per passenger
Result	74% of respondents selected satisfied or very satisfied
Other booking channels	Website, ticket offices, and travel agents

The survey team makes three statements:

1. Every passenger on our invitation list was invited.
2. We have a census of September passengers because every received response was analysed.
3. Passenger satisfaction in September was 74%.

a. Which statements can be retained in the public report, and which need revision? Give suitable wording where needed and identify the population actually described by the 74% figure.

b. Management offers funding to obtain twice as many responses next time. Recommend how the next survey should be organised and explain whether the larger response count would, by itself, support Statement 3.

Question 5 (Average)

Epsilon Retail compares two website checkout designs.

Feature	Study A	Study B
Participants	2,400 customers randomly selected from the current customer list; all participate	200 loyalty-club members who volunteer
Design received	Each customer chooses standard or simplified checkout	Each participant is randomly assigned standard or simplified checkout
Prices and delivery terms	The same under both designs	The same under both designs
Period	Both designs operate during the same month	Both designs operate during the same month
Recorded outcome	Whether the customer completes a purchase	Whether the customer completes a purchase

Each customer contributes one outcome. In both studies, the simplified checkout has the higher observed purchase-completion rate.

The marketing director prefers Study A because it includes more customers and they were randomly selected. The website manager prefers Study B because the design was randomly assigned and recommends applying its result to all customers.

a. Classify the studies and assess each manager's reasoning.

b. Which study would you rely on more for evidence that the design itself affects purchase completion? What limits remain when deciding whether to introduce it for all customers?

c. Describe a study that combines the useful features of these two designs.

Question 6 (Hard)

Zeta Manufacturing uses scheduled machine inspections. When an inspection detects wear, components can be replaced before a breakdown occurs.

The company is considering a model that predicts whether a machine will break down in the following month.

Project information	Detail
Records	10,000 machine-months from two recently opened plants, collected under the existing inspection schedule
Other plants	Older plants keep paper maintenance records, which were not included
Model development	Twelve candidate models were developed and compared using the 10,000 records
Reported result	The selected model correctly classifies 96% of those same records
Outcomes in the reported data	94% of machine-months had no breakdown
Exclusions before development	Some machine-months with confirmed breakdowns were labelled “exceptional operation” and left out; no further information about those cases is supplied

Management proposes inspecting machines predicted to have no breakdown once a month instead of twice, at every plant. Machines predicted to break down would retain twice-monthly inspections.

Recommend whether and how the company should proceed. Support your recommendation with the report and specify what evidence would change your decision.

Chapter 2: Tabular and Graphical Displays

Questions 7–16.

Report percentages to two decimal places when needed. Retain unrounded values during calculations.

Question 7 (Easy)

Eta Café records the payment method for twelve transactions. Each transaction uses one payment method.

Transaction	Method	Transaction	Method
1	Cash	7	Cash
2	Card	8	Card
3	Mobile	9	Cash
4	Card	10	Mobile
5	Cash	11	Card
6	Mobile	12	Cash

a. Construct a frequency table showing frequency, relative frequency, and percent frequency.

b. Sketch a bar chart of transaction counts, ordering the categories from most to least frequent. Include a title and a labelled numerical axis.

Question 8 (Easy)

Theta Appliances reports sales and returns for two stores. Each sale is one appliance, and each returned appliance is counted once. All sales have completed the same return window.

Store	Appliances sold	Appliances returned
Alpha	600	18
Beta	240	12

a. Calculate each store's return rate per 100 appliances sold and its share of all returned appliances.

b. Which store contributes more returned appliances, and which has the higher return rate? Explain why the rankings differ.

Question 9 (Easy)

Iota Couriers records the following delivery times:

Delivery	1	2	3	4	5	6	7	8	9	10
Time (minutes)	47	32	41	38	32	55	46	41	60	39

a. Construct a stem-and-leaf display with a leaf unit of one minute. Include a key.

b. Use the display to determine the number and percentage of deliveries taking at least 45 minutes. Identify any repeated times.

Question 10 (Average)

Kappa Services is assembling a management report. The proposed displays are listed below.

Report purpose	Available data	Proposed display
Compare payment-method shares for the quarter	Four payment categories	Pie chart
Examine the distribution of customer waiting times	Individual waiting times in minutes	Bar chart with one bar for each customer
Compare booking-channel preferences at two branches	Channel counts; 240 customers at one branch and 80 at the other	Stacked bars of customer counts
Examine the relationship between advertising expenditure and orders	One expenditure-order pair for each of eight stores	Two separate pie charts
Follow monthly sales during the year	Twelve monthly sales totals	Line chart in calendar order

Which proposed displays would you retain, replace, or supplement? For each, specify the display you would use and explain how it serves the stated purpose. More than one choice can be defensible if justified.

Question 11 (Average)

Lambda Logistics records loading times for twelve deliveries.

Delivery	1	2	3	4	5	6	7	8	9	10	11	12
Time (minutes)	18	7	30	12	25	9	34	20	3	29	10	23

a. Calculate the approximate class width for four classes using the observed range.

b. For the report, use four classes of width 10 minutes beginning at 0. Include the lower boundary and exclude the upper boundary of each class. Construct a table containing class midpoints, frequencies, percent frequencies, and cumulative percent frequencies.

c. Sketch the frequency histogram. How many deliveries took less than 20 minutes?

Question 12 (Average)

Mu Support retains only the following cumulative summary for twenty nonnegative customer waiting times. All waiting times were below 20 minutes.

Waiting-time threshold	Number below the threshold
5 minutes	4
10 minutes	11
15 minutes	17
20 minutes	20

a. Recover the frequencies and percent frequencies for the intervals $[0, 5)$, $[5, 10)$, $[10, 15)$, and $[15, 20)$.

b. The service target requires at least 80% of customers to wait less than 12 minutes. Determine whether the retained summary establishes that the target was met. Support your answer with the smallest and largest possible qualifying counts consistent with the table.

Question 13 (Average)

Nu Retail records one order from each of twelve customers.

Customer	Customer type	Order collection
1	Member	Pickup
2	Member	Pickup
3	Member	Delivery
4	Member	Pickup
5	Member	Delivery
6	Member	Pickup
7	Guest	Pickup
8	Guest	Delivery
9	Guest	Delivery
10	Guest	Delivery
11	Guest	Delivery
12	Guest	Delivery

a. Construct a crosstabulation with customer type in rows and collection method in columns. Include the marginal totals.

b. Construct separate row-percentage and column-percentage tables.

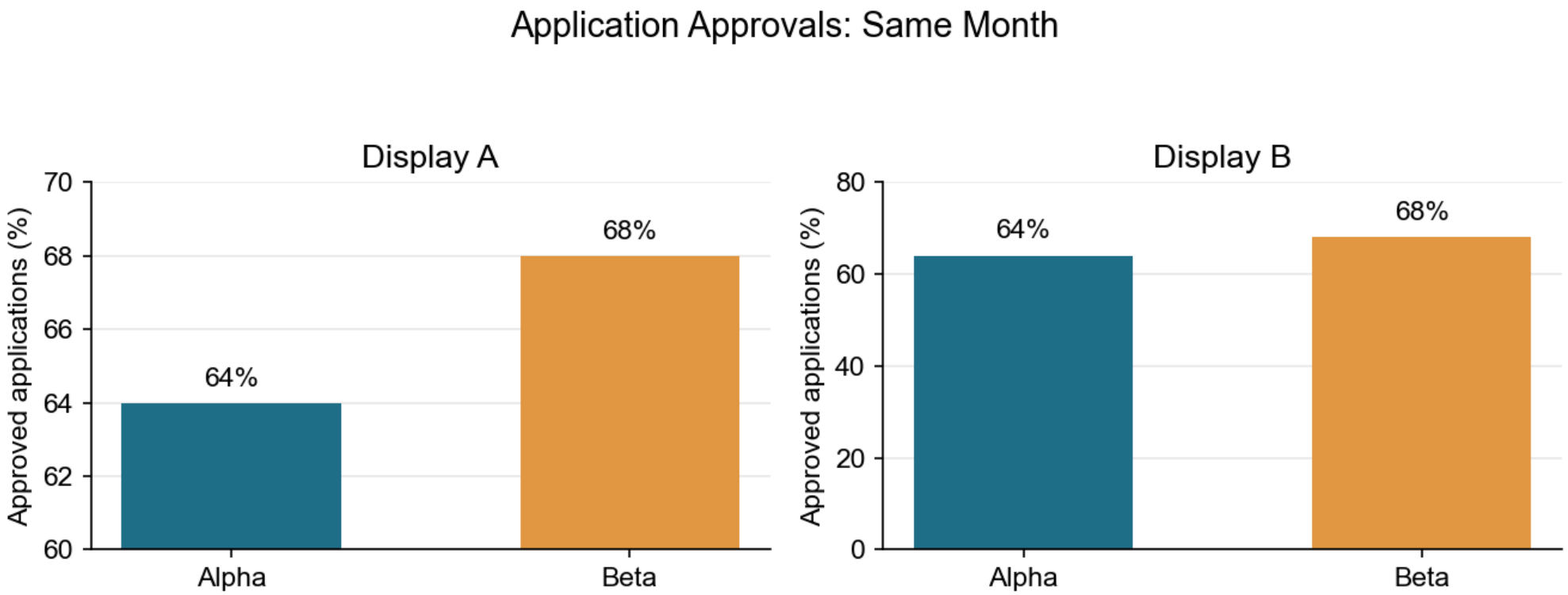
Question 13 (Average)

c. Assess the following statements using your results:

- 80% of pickup customers are members.
- 80% of members use pickup.
- One-third of all recorded customers are members who use pickup.

Question 14 (Average)

Xi Bank compares application approvals at two branches. Two displays of the same month's records are submitted for its management report.



Alpha: 160 approvals from 250 applications. Beta: 170 approvals from 250 applications.

The proposed commentary reads:

“Beta's greater approval share makes its procedure the better choice. All branches should adopt that procedure next quarter.”

Question 14 (Average)

Which display, if either, would you publish, and what commentary would you attach? Support your choices using the information supplied. No calculations are required.

Question 15 (Hard)

Omicron Maintenance compares two repair teams. Every completed job is classified as routine or complex and as on time or late against its agreed deadline.

Team	Job type	On-time jobs	Completed jobs
Alpha	Routine	18	20
Alpha	Complex	56	80
Beta	Routine	68	80
Beta	Complex	12	20

The operations director recommends Beta for a new contract, citing its overall on-time record.

Evaluate the recommendation numerically. As part of your assessment, compare the teams on a reference workload of 20 routine and 20 complex jobs each, assuming their observed within-type on-time rates apply to that workload. Explain what the records support and what remains uncertain.

Question 16 (Hard)

Pi Stores records 200 customer transactions. Every transaction is classified as promotional or regular and uses exactly one payment method.

The following figures are exact:

Report item	Value
Promotional transactions	120
Card transactions	100
Cash transactions	60
Mobile transactions	40
Promotional share of card transactions	75%
Promotional share of cash transactions	50%

A draft report makes three claims:

1. 75% of promotional transactions use cards.
2. Cards are the most common payment method within each transaction type.
3. There are more regular mobile transactions than promotional mobile transactions.

a. Reconstruct the complete transaction-type by payment-method count table, including its margins.

b. Assess all three claims with calculations. For the mobile comparison, also compare the percentage using mobile within each transaction type.

Chapter 3: Numerical Measures

Questions 17–33.

For percentiles, use $L_p = (p/100)(n + 1)$ and interpolate when the location is not an integer. Use sample variance and covariance unless the question identifies a complete population.

Retain unrounded values during calculations. Report final numerical measures to two decimal places when needed and correlation coefficients to three decimal places. State the units.

Question 17 (Easy)

Rho Repairs records the waiting times for eight customers.

Customer	1	2	3	4	5	6	7	8
Waiting time (minutes)	6	9	5	7	6	10	6	15

a. Calculate the mean, median, and mode.

b. A check of the final customer's record changes 15 minutes to 35 minutes. Recalculate the three measures and explain why they do not all respond in the same way.

Question 18 (Easy)

For a one-period portfolio exercise, use these asset returns as given.

Asset	Return
Alpha	4%
Beta	12%

Divide 10,000 MAD between the two assets at the start of the period. There are no fees or later deposits or withdrawals.

a. Find the investment amounts and portfolio weights that give a portfolio return of 9%.

b. Calculate the portfolio's monetary gain. Does the simple average of the two asset returns give this portfolio's return? Explain.

Question 19 (Easy)

Tau Couriers records processing times for nine parcels.

Parcel	1	2	3	4	5	6	7	8	9
Processing time (minutes)	22	8	14	30	10	18	12	26	16

Calculate P_{40} and P_{65}. Show the ordered data, each location, and any interpolation. Explain what a non-integer location represents.

Question 20 (Easy)

Upsilon Engineering records the following downtime for all four machines in one workshop during a month.

Machine	Alpha	Beta	Gamma	Delta
Downtime (hours)	2	4	4	6

a. Treat these four machines as the complete population of interest. Calculate the mean, population variance, and population standard deviation.

b. Instead, treat the four machines as a sample used to estimate variability across the company's larger machine fleet. Calculate the sample variance and sample standard deviation. Explain why the numerical results change although the observations do not.

Question 21 (Easy)

Phi Retail compares a day's sales at two branches with each branch's previous daily records.

Branch	Sales on the day (thousand MAD)	Previous sample mean	Previous sample standard deviation
Alpha	220	200	20
Beta	90	80	5

a. Calculate the day's z-score for each branch. Which branch has the higher sales, and which is further above its own usual level?

b. What sales figure would Alpha need to have the same z-score as Beta on this day?

Question 22 (Average)

Chi Investments reports three successive annual returns. No money is added or withdrawn.

Year	1	2	3
Return	20%	-10%	5%

a. Calculate the arithmetic mean annual return and the geometric mean annual return.

b. Starting with 1,000 MAD, calculate the ending value after the three actual returns. Which average reproduces this ending value when applied as a constant annual rate?

c. Quantify the difference in ending value if the other average is used as a constant rate.

Question 23 (Average)

Psi Distribution records delivery costs for ten shipments.

Shipment	1	2	3	4	5	6	7	8	9	10
Cost (MAD)	24	13	31	19	11	26	39	18	22	17

a. Calculate Q_1, Q_2, Q_3, and the IQR using the lecture's percentile convention.

b. An analyst states that exactly 25% of the recorded costs must be below the calculated Q_1. Assess the statement using these observations.

Question 24 (Average)

Omega Foods uses two packaging lines for different bag sizes.

Line	Sample mean net weight (g)	Sample standard deviation (g)
Alpha	40	2
Beta	80	3

The quality manager describes Alpha as more consistent. The production manager describes Beta as more consistent.

a. Calculate each line's coefficient of variation.

b. Assess both descriptions, specifying the meaning of consistency under which each could be defended.

c. Can these summaries identify which line produces a larger percentage of bags outside its permitted weight limits? Explain.

Question 25 (Average)

Alpha Dispatch records five delivery times for each of two routes.

Delivery	1	2	3	4	5
Route A (minutes)	8	9	10	11	12
Route B (minutes)	6	8	10	12	14

The contract requires delivery within 12 minutes, including delivery at exactly 12 minutes.

a. Calculate the sample mean, range, variance, and standard deviation for each route.

b. Calculate the average absolute distance from the mean for each route. Is that quantity the same as the standard deviation?

c. Compare the recorded proportions meeting the contract. Assess the claim that equal mean delivery times imply equivalent service.

Question 26 (Average)

Beta Components examines a complete batch of 10,000 items. Their processing times have population mean 50 seconds and population standard deviation 4 seconds.

a. Without assuming a distribution shape, what minimum percentage and minimum number of items can be guaranteed to have processing times in the inclusive interval 40–60 seconds?

b. Suppose a histogram of the batch is approximately bell-shaped. Estimate the number in the inclusive interval 42–58 seconds using the empirical rule.

c. A report changes the estimate in part b to “at least 9,500 items.” Assess this wording and calculate the minimum count Chebyshev's theorem guarantees for the same interval.

Question 27 (Average)

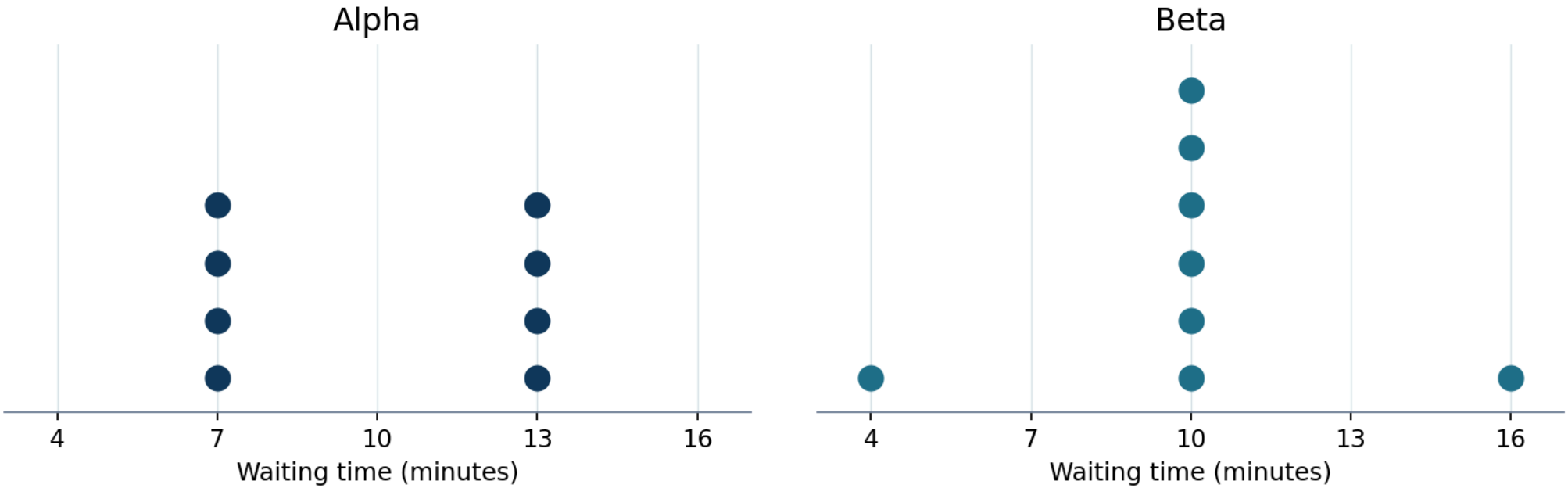
Gamma Service records the following waiting times for nine customers.

Customer	1	2	3	4	5	6	7	8	9
Waiting time (minutes)	13	8	30	11	6	16	10	14	12

Calculate the five-number summary and the IQR fences. Sketch a box plot with potential outliers shown separately. State the two observed values at which the lines extending from the box end.

Question 28 (Average)

Delta Support compares eight recorded waiting times for each of two teams.



Each dot represents one observation.

Measure	Alpha	Beta
Mean	10 minutes	10 minutes
Median	10 minutes	10 minutes
Sample standard deviation	Approximately 3.21 minutes	Approximately 3.21 minutes

Question 28 (Average)

Compare the waiting-time patterns shown in the plots. Explain what the matching summaries conceal and whether a bell-shaped description is appropriate for either set of observations. No calculations are required.

Question 29 (Average)

Epsilon Manufacturing records machine age and average output for five machines under the observed operating conditions.

Machine	Age, X (years)	Output, Y (units per minute)
Alpha	1	8
Beta	2	6
Gamma	3	7
Delta	4	3
Epsilon	5	1

Calculate the sample covariance and correlation. Show the deviations and paired cross-products in a calculation table.

State the association supported by these records. Do the results alone establish how much output would improve if an old machine were replaced?

Question 30 (Average)

Zeta Services reports a mean job cost of 25 MAD from eight jobs. A check finds that one cost was entered as 48 MAD rather than its correct value of 24 MAD. The other seven costs were recorded correctly.

A ninth job, costing 54 MAD, is then added.

a. Calculate the corrected mean for the original eight jobs and the mean after adding the ninth.

b. Can the median of the nine correct costs be determined from this information? Justify your answer.

Question 31 (Hard)

Eta Logistics records six loading delays.

Observation	1	2	3	4	5	6
Delay, X (minutes)	1	1	2	2	3	9

a. Calculate the mean, direct moment skewness, Pearson kurtosis, and excess kurtosis. Show the deviations and their second, third, and fourth powers in a calculation table.

b. The company assigns each observation a performance score $Y = 10 - X$, where X is the delay measured in minutes and higher scores are preferred. Determine the score's skewness, Pearson kurtosis, and excess kurtosis from the relationship between X and Y. Justify your results without constructing another full deviation table.

c. After switching to the score, a manager reports that the distribution's shape demonstrates an improvement in operations. Assess the claim and interpret what the calculated measures establish about these observations.

Question 32 (Hard)

Theta Repairs records waiting times for a sample of ten jobs.

Job	1	2	3	4	5	6	7	8	9	10
Waiting time (minutes)	12	9	38	11	14	10	13	12	10	11

One analyst says that 38 minutes is not unusual because it is within three sample standard deviations of the mean. Another wants it deleted from the service report as an error. The original job log has not yet been checked.

Assess both positions numerically using the z-score and IQR criteria. Quantify how retaining or omitting 38 changes the mean and sample standard deviation, and recommend what should be reported while the record is investigated.

Question 33 (Hard)

Iota Retail records advertising expenditure and orders for five stores during the same month.

Store	Advertising, X (thousand MAD)	Orders, Y
Alpha	2	40
Beta	4	120
Gamma	6	80
Delta	8	100
Epsilon	10	60

Before calculating correlation, an analyst sorts each numerical column separately from smallest to largest. The report then states:

“Advertising and orders have a perfect positive correlation. Sorting cannot change the result because it leaves each variable's mean and standard deviation unchanged.”

a. Audit this statement. Calculate the sample covariance and correlation from the original pairs and from the separately sorted columns. Show the paired calculations.

b. Using the original pairs, express advertising expenditure in MAD rather than thousand MAD. Determine the resulting covariance and correlation without treating the unit change as new observations.

c. State the conclusion about advertising and orders that the original records support.

Chapter 4: Introduction to Probability

Questions 34–50.

State the events used in probability calculations. Give exact fractions where convenient; otherwise report probabilities to four decimal places and monetary amounts to two decimal places when needed. Retain unrounded values during calculations.

Do not assume equally likely outcomes or independence unless the information supports those assumptions.

Question 34 (Easy)

Kappa Sales contacts two different customers in a specified order. Record purchase (P) or no purchase (N) for each customer. The four outcomes below exhaust the possibilities.

Outcome, first customer then second	Assigned probability
(P, P)	0.10
(P, N)	0.20
(N, P)	0.25
(N, N)	p

a. Find p and verify that the assignments form a valid probability model.

b. List the outcomes in each event and calculate its probability: the first customer purchases; exactly one customer purchases; at least one customer purchases.

c. An assistant assigns probability $1/4$ to each outcome because there are four rows. Assess that assignment.

Question 35 (Easy)

Lambda Delivery sells the following service configurations.

Service	Available dispatch days	Available parcel sizes
Economy	Monday, Wednesday, Friday	Small, large
Priority	Tuesday, Thursday	Small, medium, large

For either service, the customer also chooses prepaid or invoice billing. Every combination within a service is available. A configuration specifies the service, dispatch day, parcel size, and billing method.

a. How many configurations are available altogether?

b. How many use invoice billing and a large parcel?

c. A catalogue designer multiplies 2 services by 5 days by 3 sizes by 2 billing methods and reports 60 configurations. Explain the error.

Question 36 (Easy)

Mu Consulting has six distinct employees. Four, including Alpha, are certified to lead an assignment; two are not. Any employee can act as reviewer.

Each assignment needs one lead and one reviewer, and the same employee cannot hold both roles.

a. How many lead–reviewer assignments would be possible without the certification requirement?

b. How many meet the certification requirement?

c. How many valid assignments give Alpha one of the two roles?

Question 37 (Easy)

Nu Investments will fund three distinct projects from the following pool.

Project category	Number available
Energy	3
Logistics	4
Total	7

Every three-project group is affordable and equally likely to be selected. A project cannot be selected twice, and the order of selection is ignored.

a. How many groups are possible?

b. How many contain exactly two energy projects? What is the probability of selecting such a group?

c. Calculate the probability that the selected group contains at least one energy project.

Question 38 (Easy)

Xi Installations reviews 80 completed contracts.

Completion	Rework required	No rework	Total
Late	6	14	20
On time	10	50	60
Total	16	64	80

Select one of these contracts uniformly at random. Let L denote late completion and R denote rework.

Calculate the probability that the contract:

a. Was late.

b. Was late and required rework.

c. Had at least one of the two problems.

d. Had exactly one of the two problems.

e. Had neither problem.

Question 39 (Average)

Omicron Claims has eight distinct claim files: three flagged and five unflagged. Two files are selected sequentially, uniformly at each draw, without replacement.

a. Calculate the probability that both selected files are flagged.

b. Calculate the probability that exactly one selected file is flagged.

c. Given that at least one selected file is flagged, calculate the probability that exactly one is flagged.

d. If the first file is replaced before the second draw, what is the probability that both selections are flagged? Compare it with part a.

Question 40 (Average)

Pi Freight records the following 80 shipments.

Destination	Insured	Not insured	Total
Domestic	30	18	48
International	20	12	32
Total	50	30	80

Select one recorded shipment uniformly at random. Let I denote international and C denote insured.

a. Calculate $P(I \mid C)$ and $P(C \mid I)$ and interpret each.

b. Determine whether I and C are independent in these records. Support the answer numerically.

c. Are the events mutually exclusive? Explain how that answer can be consistent with part b.

Question 41 (Average)

Rho Retail receives a probability report about individual orders.

Event	Reported probability
A: customer uses the app	0.60
B: customer uses a discount	0.55
Both A and B	0.10

a. Can all three probabilities describe the same order population? Justify your answer numerically.

b. Keep $P(A) = 0.60$ and $P(B) = 0.55$ unchanged. Determine the full range of possible values for $P(A \cap B)$.

c. If A and B were independent, calculate the probability of both, neither, and exactly one event.

Question 42 (Average)

Sigma Hotels receives bookings through three channels. Each booking belongs to exactly one channel.

Channel	Current share of bookings	Cancellation probability within channel	Proposed share of bookings
App	0.50	0.04	0.20
Website	0.30	0.10	0.30
Agent	0.20	0.15	0.50

The proposed shares sum to 1. Assume the channel-specific cancellation probabilities would remain unchanged.

a. Calculate the overall cancellation probability under the current shares.

b. Calculate it under the proposed shares.

c. Quantify the change in percentage points. Would that change establish that cancellation behaviour had worsened within a channel?

Question 43 (Average)

Tau Payments uses a primary connection and a backup connection. Transactions can be processed whenever at least one connection is working.

Connection	Probability of working during the period
Primary	0.90
Backup	0.80

Initially assume the working statuses are independent.

a. Calculate the probability that transactions can be processed and the probability that exactly one connection is working.

b. Given that transactions can be processed, calculate the probability that the primary is down and the backup is carrying the service.

c. During heat alerts, the two individual working probabilities remain as above, but the probability that both connections are down is 0.07. Calculate service availability during a heat alert and assess whether the independence assumption remains valid.

Question 44 (Average)

Upsilon Assembly receives components from three suppliers. Each component comes from exactly one supplier.

Supplier	Share of components	Defect probability
Alpha	0.60	0.02
Beta	0.30	0.03
Gamma	0.10	0.09

a. Calculate the overall defect probability.

b. Given that a randomly selected component is defective, calculate the probability of each supplier being its source and rank the suppliers.

c. Compare this ranking with the ranking by defect probability within each supplier. Explain the difference.

Question 45 (Average)

Phi Commerce screens orders for fraud.

Quantity	Probability
An order is fraudulent	0.02
Flag given a fraudulent order	0.90
Flag given a legitimate order	0.06

a. Calculate the probability that an order is flagged and the probability that a flagged order is fraudulent.

b. Calculate the probability that an unflagged order is fraudulent.

c. In a different order population, the fraud probability is 0.10, while both conditional flag probabilities remain unchanged. Recalculate the probability of fraud given a flag. Explain why it changes.

Question 46 (Average)

Chi Logistics needs a probability of late arrival for shipments on a newly introduced route.

Assessment	Evidence and reasoning
Operations analyst: 0.10	Ten percent of 10,000 shipments were late on the old route last year. The new route uses a different port and has a shorter connection window.
Route manager: 0.25	Judgement based on recent port reports and experience with similar connection windows. No matching shipment dataset or record of previous forecast accuracy is supplied.

The director wants to adopt 0.10 solely because it comes from many recorded shipments, describing 0.25 as “only an opinion.”

What would you advise? Explain how each probability was assigned, what each assessment does and does not establish, and what evidence would support the number used for the new route. No calculations are required.

Question 47 (Hard)

Psi Engineering forms a four-person project team from nine distinct employees.

Qualification	Employees
Senior	Alpha, Beta, Gamma, Delta
Junior	Epsilon, Zeta, Eta, Theta, Iota

The team must contain at least two seniors. Alpha and Beta cannot serve together.

A complete appointment names the four team members, one senior team member as lead, and a different team member as recorder. Any selected member other than the lead can be recorder.

a. Count the eligible four-person teams.

b. Count the complete appointments.

c. Compare the probability of obtaining a team with exactly three seniors under these two procedures:

- Select an eligible team uniformly, then choose its lead uniformly from its seniors and its recorder uniformly from the other members.
- Select uniformly from all complete appointments.

Are the two procedures equivalent for choosing the team's composition?

Question 48 (Hard)

Omega Components applies two tests to the same component.

Actual component status	Population share	Probability Test 1 flags	Probability Test 2 flags
Defective	0.05	0.80	0.80
Acceptable	0.95	0.10	0.10

Within either actual-status group, the two test results are independent: knowing one test's result does not change the other test's flag probability within that group.

a. Calculate the probability that both tests flag a component and the probability that such a component is defective.

b. Calculate the probability that exactly one test flags and the probability of a defect given exactly one flag.

c. An analyst treats the two flags as independent without conditioning on actual status. Check that claim numerically.

Question 49 (Hard)

Alpha Marketplace considers reviewing every order flagged by its fraud system.

Quantity	Value
Orders in each period	10,000
Fraud probability in the usual period	0.03
Fraud probability in a low-fraud period	0.002
Flag probability given fraud, in either period	0.80
Flag probability given a legitimate order, in either period	0.05
Review cost per flagged order	12 MAD
Loss from each fraudulent order not reviewed	150 MAD

A review identifies the order's true status and prevents the entire fraud loss if the order is fraudulent. Legitimate orders cause no fraud loss. Unflagged orders are not reviewed, there is no review-capacity limit, and no other costs apply.

Recommend whether to review every flagged order in each period. Support the recommendation with expected counts and total costs compared with reviewing no orders. Also determine the review cost per flagged order at which the two policies have equal expected total cost in the usual period.

Question 50 (Hard)

Beta Distribution has two batches of distinct components.

Batch	Defective	Acceptable	Total
Alpha	2	4	6
Gamma	1	3	4

Two sampling procedures are proposed:

- Procedure A: choose Alpha with probability 0.60 or Gamma with probability 0.40, then select two components uniformly without replacement from that chosen batch.
- Procedure B: pool all ten components, then select two uniformly without replacement from the pool.

Under Procedure A:

a. Calculate the probability that exactly one selected component is defective.

b. Given exactly one defect in the selected pair, calculate the probability that the pair came from Alpha.

c. Given that the selected pair contains at least one defect, calculate the probability that both components are defective.

Then calculate the probability of exactly one defect under Procedure B. A colleague argues that the procedures must give the same result because the batch-selection probabilities in A match the batches' shares of the ten components. Assess that argument.